# Shounak Kamutala

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.28 |
| Mahatma Gandhi Junior College | Class XII (Maharashtra State ) | 2023 | 85.00% |
| Queen's School | Class X (CBSE) | 2021 | 95.20% |

## TECHNICAL SKILLS

- **Programming Languages**: Python, SQL
- **Data Structures and Algorithms:** Arrays, Strings, Hashing, Sliding Window, Binary Search, Stacks, Queues, Linked Lists, Trees, Graphs, Dynamic Programming, Dijkstra's Algorithm
- **Computer Science**: Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **Developer Tools:** Git, GitHub, VS Code, Google Colab

## CODING PROFILE

- **Leetcode :** 450+ problems solved | Profile

## PROJECTS

**Movie Recommender System | Python, Pandas, Scikit-learn, Streamlit, TMDB API**

- Built a **content-based** movie recommender using **CountVectorizer** and **cosine similarity** in **Scikit-learn** to suggest the five most similar movies for any selected title.
- Integrated the **TMDB API** with **multithreaded requests**, retries, and caching to fetch posters reliably, and built an interactive **Streamlit** interface with a searchable, paginated catalogue browser.
- Cut the similarity matrix from **176 MB to 46 MB** using **float16 quantisation**, secured the API key with **Streamlit secrets**, and managed the project on **GitHub**.

**DSA Visualizer | Python, Tkinter, Data Structures & Algorithms, Pytest, GitHub Actions**

- Developed an interactive **step-by-step visualizer** for 20+ data-structure and algorithm routines covering **sorting, trees, heaps, and graphs**, with pseudocode and algorithm-complexity displays.
- Implemented **BFS, DFS, Dijkstra, A*, AVL trees, BST operations, heaps, and six sorting algorithms**, with playback controls, race mode, and GIF export.
- Built a modular, testable architecture with **Pytest unit tests and GitHub Actions CI**, automating linting and testing across multiple Python versions.

## POSITIONS OF RESPONSIBILITY

**Table Tennis Coordinator**

- **Organized** and **managed** multiple **table tennis events**, coordinating logistics, scheduling, and participant operations to ensure smooth execution.
- **Promoted inter-collegiate table tennis competitions**, driving student participation and increasing visibility of competitive TT events across colleges.

## RELEVANT COURSES

- **Computer Science and AI**: Introduction to Programming, Artificial Intelligence, Data Structures and Applications
- **Mathematics**: Calculus I, Calculus II, Elementary Linear Algebra, Differential Equations
- **Statistics**: Introduction to Statistics, Probability
- **Physics:** Maths for Physics, Advanced Mathematical Physics, Computational Physics, Optics and Photonics, Fluid

Dynamics, Special Relativity, Classical Mechanics, Electronic Device Physics
- **Other**: Communication Skills, Introduction to Entrepreneurship, Macroeconomics

## EXTRACURRICULAR

- **Sports:** Represented the hostel and department in inter-college Table Tennis tournaments.
- **NSS Volunteer:** Participated in NSS activities.